

Python For Data Science (3150713)

Dmart Grocery Product Sales Analysis & Prediction

(Course-Integrated Practical & Analytical Task (CIPAT))

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CERTIFICATE

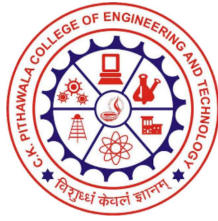
This is to certify that project work embodied in this report entitled **Demart Grocery Sales Analysis and Prediction** was carried out by **Parekh Khantkumar (230090107112), Afinwala Hitesh (230090107002)** at **C. K. Pithawala College of Engineering and Technology, Surat** as a part of **Course-Integrated Practical & Analytical Task (CIPAT)** for subject **Python For Data Science (3150713)**. This project work has been carried out under my supervision.

Date :

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DECLARATION

We hereby certify that We are the sole author of this report and that neither any part of this work nor the whole of the work has been submitted for a degree to any other University or Institution.

We declare that this is a true copy of our report, including any final revisions, as approved by my supervisor.

Date : 22 June 2025

Place : Surat, Gujarat

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Table of Contents

Title page	i
Certificate	ii
Declaration	iii
Acknowledgements	iv
Table of Contents	v
List of Figures	vi
List of Tables	vii
Abstract	viii
1 Project Introduction	1
2 Literature Review / Background	3
3 Problem Definition	6
4 Objectives	8
5 Methodology / System Design	9
6 Implementation	13
7 Results and Discussion	17
8 Conclusion and FutureWork	24
9 References	26

List of Figures

Figure 5.1 Workflow Diagram	8
Figure 7.1 EDA - Category wise sale	17
Figure 7.2 EDA - Sub Category wise sale	18
Figure 7.3 EDA -Yearly Trend	19
Figure 7.4 Correlation Matrix	19
Figure 7.5 Comparison Of Actual and Predicted Sale	20
Figure 7.6 Comparison Of Actual and Predicted Profit	21
Figure 7.7 Prediction Result of Year 2019	22
Figure 7.8 UI- Analysis Tab	23
Figure 7.9 UI- Prediction Tab	23

Dmart Grocery Product Sales Analysis and Prediction

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Abstract

Retail businesses today generate massive amounts of data through their daily sales operations. Analyzing and forecasting sales trends helps companies make data-driven decisions that lead to higher profitability and efficiency.

This project focuses on the analysis and prediction of sales and profit data for DMart grocery stores located in Tamil Nadu. The dataset was obtained from Kaggle and contains information on product categories, subcategories, sales, discounts, profit, and regional distribution.

The main objective of this study is to identify the most profitable product categories and forecast sales and profit for the upcoming year using machine learning techniques. Data preprocessing, feature extraction, and exploratory data analysis (EDA) were performed to uncover important trends. A Random Forest Regression model was then trained to predict future sales and profits for each product category and subcategory.

1 PROJECT INTRODUCTION

1.1 Background of the problem

In the modern retail landscape, companies continuously seek innovative methods to remain competitive. The rise of big data analytics and machine learning has revolutionized how organizations interpret market behavior, manage supply chains, and plan business strategies. Retail stores like DMart handle millions of transactions daily, resulting in extensive datasets containing customer information, product details, discounts, and regional performance metrics.

Traditional methods of decision-making often relied on intuition or limited reports. However, with the help of data analytics, it is now possible to identify meaningful trends hidden within massive datasets. Predictive analytics, in particular, allows organizations to forecast future sales, understand seasonal variations, and optimize inventory based on demand.

In this context, analyzing DMart's grocery sales data offers valuable opportunities to understand which products perform best, which regions generate maximum profit, and how to adjust future strategies to improve revenue growth.

The focus of this study is to perform detailed analysis and build a predictive model that forecasts future sales and profits based on historical patterns.

1.2 Importance in Real-World Scenario

DMart, founded by Mr. Radhakishan Damani in 2002, is one of India's most successful retail chains. It operates on a customer-centric business model focusing on offering quality products at affordable prices. The company's stores are spread across various regions in India, providing daily essentials such as groceries, dairy products, packaged foods, and household goods.

Over the years, DMart has built a strong reputation for offering discounts and value-based pricing. However, with hundreds of products and fluctuating demand patterns, the challenge lies in effectively managing inventory, setting discounts, and forecasting sales.

By applying data analytics and predictive modeling to DMart's grocery sales data, this project seeks to provide actionable recommendations that align with the company's strategic goals.

Importance of Analysis:

The significance of this study lies in the combination of data analytics and machine learning to address real-world retail challenges. Some of the major benefits of sales prediction include:

- **Efficient Inventory Management:** Forecasting helps maintain optimal stock levels to meet demand without overstocking.
- **Better Marketing Strategies:** Identifying high-performing categories allows targeted promotions and campaigns.
- **Profit Maximization:** Focusing on top-performing categories increases overall profitability.
- **Customer Satisfaction:** Ensuring product availability improves the customer experience.

By using predictive analytics, DMart can make more informed and proactive business decisions rather than reactive ones.

1.4 Scope of the Project

The project focuses exclusively on DMart stores located in Tamil Nadu, covering grocery products under multiple categories and subcategories. The study aims to:

- Explore the structure and content of the dataset.
- Identify the most profitable and high-demand product categories.
- Build a predictive model to forecast future sales and profit.
- Visualize insights through charts and graphs for better understanding.
- Provide data-driven recommendations to management.

The study does not include customer demographic data or external market factors such as competitor pricing or seasonal demand fluctuations, as they were not available in the dataset.

2 LITERATURE REVIEW / BACKGROUND

This section reviews previous studies and existing techniques related to retail sales prediction and data analytics. It highlights how researchers have used statistical and machine learning models to forecast sales, analyze customer trends, and improve business decisions. The review also identifies gaps in past research and explains how this project applies the Random Forest algorithm to address those challenges in the context of DMart's grocery sales data.

2.1 Machine Learning Approaches in Retail Sales Forecasting

Machine learning plays a very important role in predicting retail sales. Unlike traditional methods that depend on simple equations, machine learning models can study large datasets and automatically find hidden patterns between different factors such as product type, region, discount, and profit.

Common algorithms used in sales forecasting include:

- **Decision Trees:** easy to understand but may overfit on small datasets.
- **Random Forest:** combines many trees to give more accurate and stable results.
- **Gradient Boosting / XGBoost:** strong models that improve results by reducing errors step by step.
- **Neural Networks:** learn complex relationships but need large data and high computing power.

For this project, Random Forest Regression is used because it:

- Handles both numerical (sales, profit) and categorical (category, region) data.
- Reduces overfitting and provides high prediction accuracy.
- Shows which features affect sales the most through “feature importance.”

This makes Random Forest ideal for DMart's dataset, which includes a mix of product categories and regional sales data.

2.2 Data Preprocessing and Visualization

Before applying any machine learning model, the dataset must be properly cleaned and prepared. This process is called data preprocessing. It ensures that the model learns from accurate and complete information.

Main preprocessing steps used in this project:

- Handling missing values: Filling or removing records with incomplete information.
- Encoding categorical data: Changing text data (like region names or categories) into numbers using Label Encoding or One-Hot Encoding.
- Scaling and normalization: Keeping all numeric data (sales, profit, discount) on a similar scale to improve model performance.

After preprocessing, data visualization helps in understanding patterns and trends in the dataset. Libraries like Matplotlib and Seaborn are used to create charts and graphs.

Examples include:

- Bar charts showing which categories have the highest sales.
- Pie charts displaying regional profit contributions.
- Heatmaps showing relationships between sales range and product frequency.

Visualization makes it easier to explain findings and helps managers see which product categories perform best.

2.3 Relevance to the Project

The techniques and studies discussed in previous research are directly applied in this DMart Grocery Sales Analysis and Prediction project. Many earlier works focused on one product or one region, but this project looks at multiple categories and regions to give a broader view of sales patterns.

The chosen model, Random Forest Regression, fits the project's requirements because it:

- Works well with both categorical and numerical data.
- Provides clear, interpretable results.

- Performs efficiently even with medium-sized datasets.

This project also uses easy-to-access tools such as Python, Pandas, NumPy, Scikit-learn, and Seaborn, making it suitable for academic study and small-scale business applications.

By applying proven machine learning methods to real-world DMart data, this project helps identify:

- Which categories bring the most sales and profit.
- How discounts and regions affect performance.
- How sales might grow in the following year.

Hence, this study connects theoretical knowledge with practical business insights, showing how data analytics can support smart decision-making in retail.

3 PROBLEM DEFINITION

In today's competitive retail market, businesses generate large amounts of sales and customer data every day. However, much of this data remains underutilized without proper analysis. Retailers like DMart need to understand their data to make informed decisions about which products sell best, which regions are most profitable, and how discounts impact sales and profit margins.

Predictive analytics and machine learning provide powerful tools to extract meaningful insights from this data. By studying past sales patterns, a model can forecast future sales and profits, helping management plan inventory, pricing, and promotions more effectively.

DMart operates multiple stores across different cities and regions, selling thousands of grocery products every day. Each product has unique sales patterns influenced by factors such as:

- Product category and subcategory,
- Region or city of the store,
- Discounts applied on products, and
- Seasonal customer demand.

As a result, DMart's management faces several challenges in understanding:

- Which product categories generate the most revenue.
- How much profit each region contributes.
- How to forecast next year's sales to manage stock and promotions efficiently.

Without a proper data-driven approach, these decisions often rely on assumptions rather than evidence, leading to losses or missed opportunities.

Challenges in the Problem

Several challenges arise while analyzing and predicting retail sales data:

- **Large and diverse data:** The dataset contains multiple product categories and cities, requiring proper handling of categorical and numerical data.

- **Data preprocessing:** Missing values, inconsistent formats, and duplicate entries must be corrected before analysis.
- **Feature selection:** Identifying which factors (e.g., discount, category, region) most influence sales and profit is crucial.
- **Model selection and tuning:** Choosing a model that balances accuracy and interpretability, such as Random Forest Regression.
- **Visualization and interpretation:** Making complex data understandable to business users through charts and reports.

4 OBJECTIVE

Every project begins with a clear set of objectives that define what the research aims to achieve. In this project, the main goal is to use data analytics and machine learning techniques to analyze DMart's grocery sales data and forecast future sales and profits.

Our main objectives are mentioned below.

Main Objective:

- To collect and understand DMart's grocery sales dataset from Kaggle, including attributes like Category, Region, Discount, Sales, and Profit.
- To clean and preprocess the data by handling missing values, removing duplicates, encoding categorical features, and normalizing numerical columns.
- To perform Exploratory Data Analysis (EDA) to study sales and profit distribution across categories and regions using Python visualization libraries.
- To apply feature engineering by creating new features like Year, Month, and Profit Margin and selecting the most relevant predictors.
- To build a Random Forest Regression model for predicting DMart's future sales and profit based on historical data.
- To evaluate model performance using metrics like R^2 Score and Mean Absolute Error (MAE) for accuracy verification.
- To forecast 2019 sales and profit, generating category-wise and region-wise predictions for better planning.
- To generate insights and recommendations that help DMart improve stock management, discount strategies, and profitability.

5 METHODOLOGY / SYSTEM DESIGN

The methodology describes the step-by-step process used to complete the project. It explains how the dataset was collected, processed, analyzed, and used to build and test the prediction model. The system is designed to ensure accurate results, proper data flow, and efficient use of machine learning techniques.

5.1 System Overview

The system follows a structured approach starting from data collection to prediction. The main stages of the process are:

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Engineering and Selection
5. Model Building & Model Evaluation
6. Prediction and Insights

This stepwise structure ensures smooth workflow and reliable output.

Workflow Diagram:

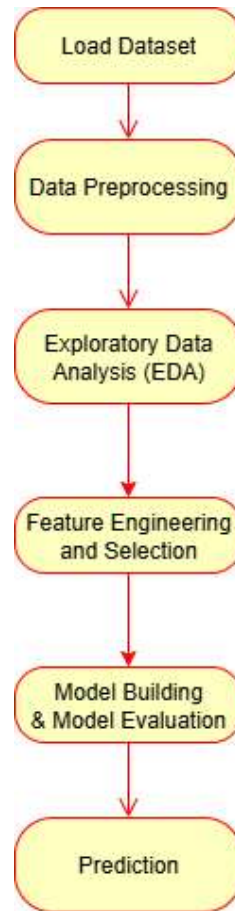


Fig 5.1: Workflow

5.2 Data Collection

The dataset for this project is taken from Kaggle, containing DMart's grocery sales information. It includes attributes such as Category, Subcategory, Region, Discount, Sales, and Profit.

The data covers the years 2015 to 2018, which are used to forecast the sales and profit for 2019. This dataset provides a strong foundation for retail analysis and prediction.

5.3 Data Preprocessing

Before analysis, the raw dataset must be cleaned and prepared.

- Missing values are handled using mean or median replacement.
- Duplicate or irrelevant records are removed.
- Categorical variables such as Category, City, and Region are converted into numeric form using Label Encoding or One-Hot Encoding.
- Numerical columns like Sales and Profit are normalized to keep all features on a similar scale.

After preprocessing, the dataset becomes clean, consistent, and ready for analysis.

5.4 Exploratory Data Analysis (EDA)

EDA is performed to understand data behavior and identify patterns.

- The distribution of sales and profit is analyzed for each category and region.
- Top-performing and low-performing products are identified.
- Data visualization tools such as Matplotlib and Seaborn are used to create bar charts, pie charts, and heatmaps that show trends clearly.

This step helps to uncover relationships between discounts, sales, and profits.

5.5 Feature Engineering and Selection

New features are created and the most relevant ones are selected to improve model performance.

- New features such as Year, Month, and Profit Margin are added to capture time-based trends.
- Correlation analysis and feature importance from Random Forest are used to identify the most significant variables.

These selected features ensure better accuracy in prediction and make the model more interpretable.

5.6 Model Building & Model Evaluation

After preparing the data, the model is built using Random Forest Regression, which is an ensemble learning technique.

- The data is divided into training (80%) and testing (20%) parts.
- The Random Forest model is trained on the training data.
- Model parameters like the number of trees and maximum depth are tuned to achieve better performance.

Once the model is trained, its performance is evaluated to ensure accuracy and reliability:

- Metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) are used to measure performance.
- Predicted results are compared with actual data using graphs and error plots. Evaluation helps verify that the model provides consistent and realistic predictions.

5.7 Prediction and Forecasting

The trained model is then used to predict future sales and profits.

- The model forecasts 2019 sales and profit using the historical data from 2015 to 2018.
- Predictions are generated for each product category and region. This step helps DMart understand expected sales growth and profit trends for the coming period.

6 IMPLEMENTATION

The implementation phase focuses on putting the planned methodology into action. It involves importing the required tools, performing data cleaning, analyzing the dataset, building the machine learning model, and finally evaluating and predicting sales and profits. Each step was carefully executed to ensure accuracy and practical usability of the results.

6.1 Libraries used

The following libraries were used:

- Pandas – for data loading, cleaning, and manipulation.
- NumPy – for performing mathematical and numerical operations.
- Matplotlib – for creating graphs and visualizing results.
- Seaborn – for advanced and more aesthetic data visualization.
- Scikit-learn – for implementing machine learning algorithms such as Random Forest Regression and evaluating performance

6.2 Data Loading

The dataset, DMart Grocery Sales Data, was imported from Kaggle using Pandas. The data contained records for multiple product categories, regions, discounts, and profits for the years 2015–2018.

Code example:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("DMart_Grocery_Sales_-_Retail_Analytics_Dataset.csv")

df
```

6.3 Data Cleaning

The dataset is already cleaned so there is no need for cleaning of the data set.

6.4 Exploratory Data Analysis (EDA)

Since the data was already clean, the focus was on exploring patterns and relationships among different features.

Using Matplotlib and Seaborn, several visualizations were created to understand sales behavior and business performance.

Code example:

```
import seaborn as sns

import matplotlib.pyplot as plt

sns.barplot(x='Category', y='Sales', data=data)

plt.title("Category-wise Sales Comparison")

plt.show()
```

6.5 Feature Engineering

To improve prediction performance, a few analytical columns were added to the dataset.

For instance, a Profit Margin column was derived to represent the ratio between profit and sales:

```
data['Profit_Margin'] = (data['Profit'] / data['Sales']) * 100
```

6.6 Model Training

The Random Forest Regressor algorithm was used to build the prediction model.

It was chosen because it handles mixed data types effectively and provides high accuracy for regression tasks.

Code example:

```
# Step 1: Feature Preparation

df_model = df.copy()

# Month is already numeric

features = ['Year', 'Month', 'Category', 'Sub Category']
```

```

categorical_cols = ['Category', 'Sub Category']

numerical_cols = ['Year', 'Month']

# One-Hot Encode categorical columns

encoder = OneHotEncoder(sparse_output=False, drop='first')

encoded_cats = encoder.fit_transform(df_model[categorical_cols])

encoded_df = pd.DataFrame(encoded_cats,
columns=encoder.get_feature_names_out(categorical_cols))

X = pd.concat([df_model[numerical_cols].reset_index(drop=True), encoded_df], axis=1)

y_sales = df_model['Sales']

y_profit = df_model['Profit']

```

Step 2: Train/Test Split for Evaluation

```

X_train, X_test, y_sales_train, y_sales_test, y_profit_train, y_profit_test =
train_test_split(
    X, y_sales, y_profit, test_size=0.2, random_state=42
)

```

Step 3: Train Random Forest Models

```

rf_sales = RandomForestRegressor(n_estimators=500, random_state=42)

rf_profit = RandomForestRegressor(n_estimators=500, random_state=42)

rf_sales.fit(X_train, y_sales_train)

rf_profit.fit(X_train, y_profit_train)

```

6.7 Forecasting Future Sales and Profit

After evaluation, the trained model was used to predict 2019 sales and profit based on past patterns (2015–2018).

Code example:

```
# Step 4: Evaluate Models on Test Set
```

```
sales_pred_test = rf_sales.predict(X_test)
```

```
profit_pred_test = rf_profit.predict(X_test)
```

```
# Step 5: Predict Next Year
```

```
future_year = df_model['Year'].max() + 1
```

```
future_data = df_model.copy()
```

```
future_data['Year'] = future_year
```

```
# Encode categorical columns for future data
```

```
encoded_future = encoder.transform(future_data[categorical_cols])
```

```
encoded_future_df = pd.DataFrame(encoded_future,  
columns=encoder.get_feature_names_out(categorical_cols))
```

```
future_X = pd.concat([future_data[numerical_cols].reset_index(drop=True),  
encoded_future_df], axis=1)
```

```
future_data['Predicted_Sales'] = rf_sales.predict(future_X)
```

```
future_data['Predicted_Profit'] = rf_profit.predict(future_X)
```

```
# Step 6: Group Predictions by Category & Sub-Category
```

```
future_summary = future_data.groupby(['Category', 'Sub  
Category'])[['Predicted_Sales', 'Predicted_Profit']].sum().reset_index()
```

```
print("\n-----Next Year Predicted Sales & Profit-----")
```

```
print(future_summary[future_summary['Predicted_Sales'] > 0])
```

7 RESULTS AND DISCUSSION

This section presents the outcomes of the DMart Grocery Sales and Profit Prediction project. It includes exploratory data analysis (EDA), correlation insights, model performance visualization, and predicted results for 2019. The aim is to evaluate model accuracy and highlight business insights obtained through machine learning.

7.1 Exploratory Data Analysis (EDA)

Here we have mentioned some of the visualization of our dataset.

1. Category-wise Sales

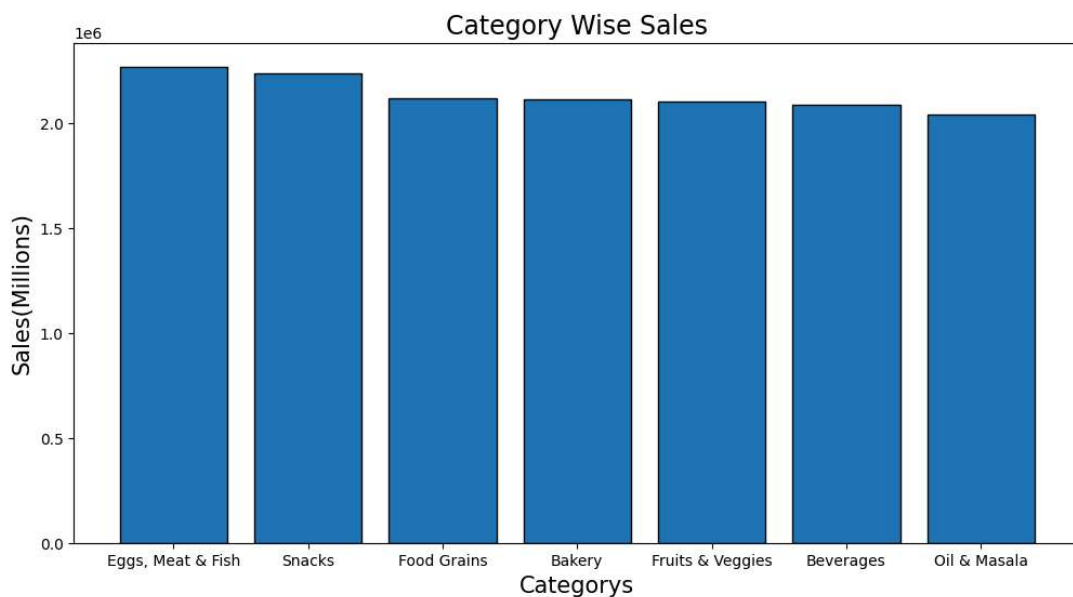


Fig 7.1: EDA-Category-wise Sales

Observation:

- Eggs, Meat & Fish, Snacks, and Food Grains recorded the highest total sales.
- Oil & Masala contributed relatively lower sales compared to other categories.
- The sales distribution was fairly consistent across major categories, indicating balanced inventory management.

2. Subcategory-wise Sales

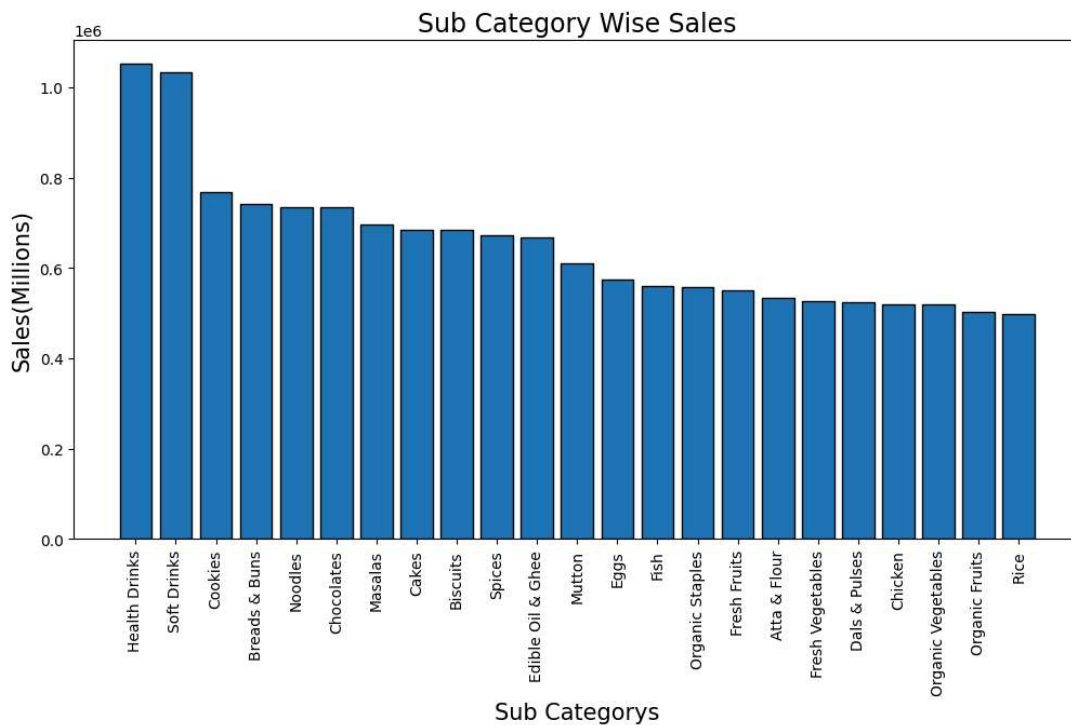


Fig 7.2: EDA-Sub Category-wise Sales

Observation:

- Health Drinks and Soft Drinks were the leading subcategories.
- Other strong performers included Cookies, Breads & Buns, and Noodles.
- Lower-performing subcategories such as Organic Vegetables and Rice show potential for targeted promotions or improved marketing.

3. Yearly Sales and Profit Trend

Observation:

- Sales showed a continuous upward trend each year, growing sharply between 2017 and 2018.
- Profits also increased steadily, indicating improved operational efficiency and cost control.
- The overall trend shows DMart's strong growth trajectory over the analyzed period.

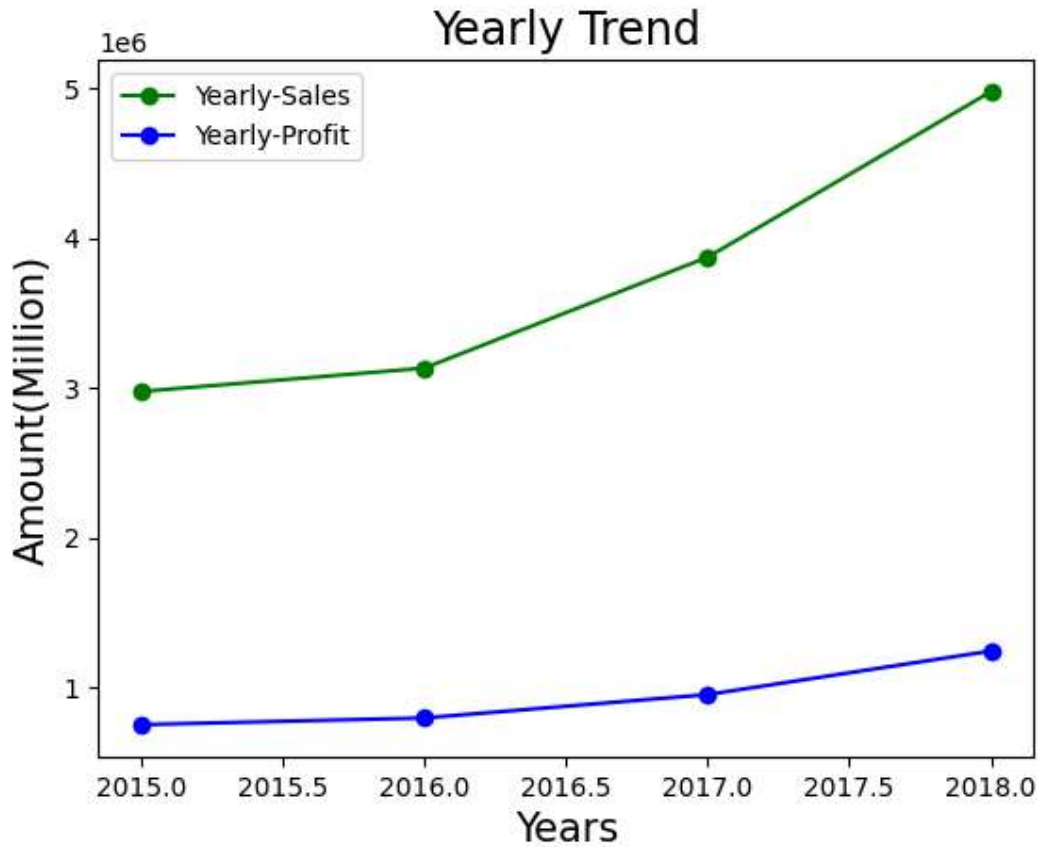


Fig 7.3: EDA-Yearly Sales and Profit Trend

7.2 Correlation Analysis

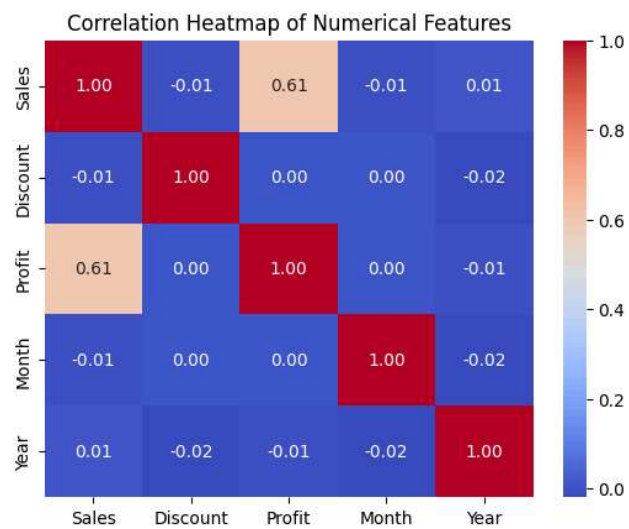


Fig 7.4: Correlation Matrix

Observation:

- Sales and Profit had a strong positive correlation (0.61), confirming that higher sales generally lead to higher profit.
- Discount showed a weak correlation with profit, implying that discounts impact profit differently across categories.
- Month and Year were weakly correlated with sales, suggesting consistent performance across time periods.

7.3 Model Prediction Results

The Random Forest Regression Model was trained to predict both Sales and Profit. The comparison between actual and predicted values was visualized using scatter plots.

1. Comparison of Actual vs Predicted Sales

- The plotted points are closely aligned with the red diagonal “perfect prediction” line, indicating that the model achieved high accuracy in predicting sales.
- The results show minimal deviation between actual and predicted values.

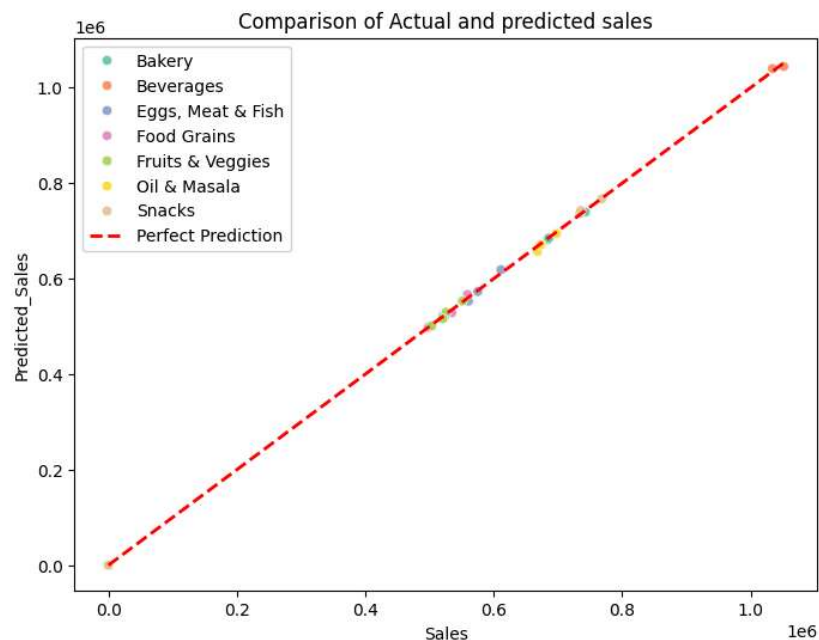


Fig 7.5: Comparison of Actual vs Predicted Sales

2. Comparison of Actual vs Predicted Profit

- Similar to the sales plot, the profit prediction also matched closely with the actual data.
- This confirms that the model effectively learned relationships between factors like discount, category, and region.

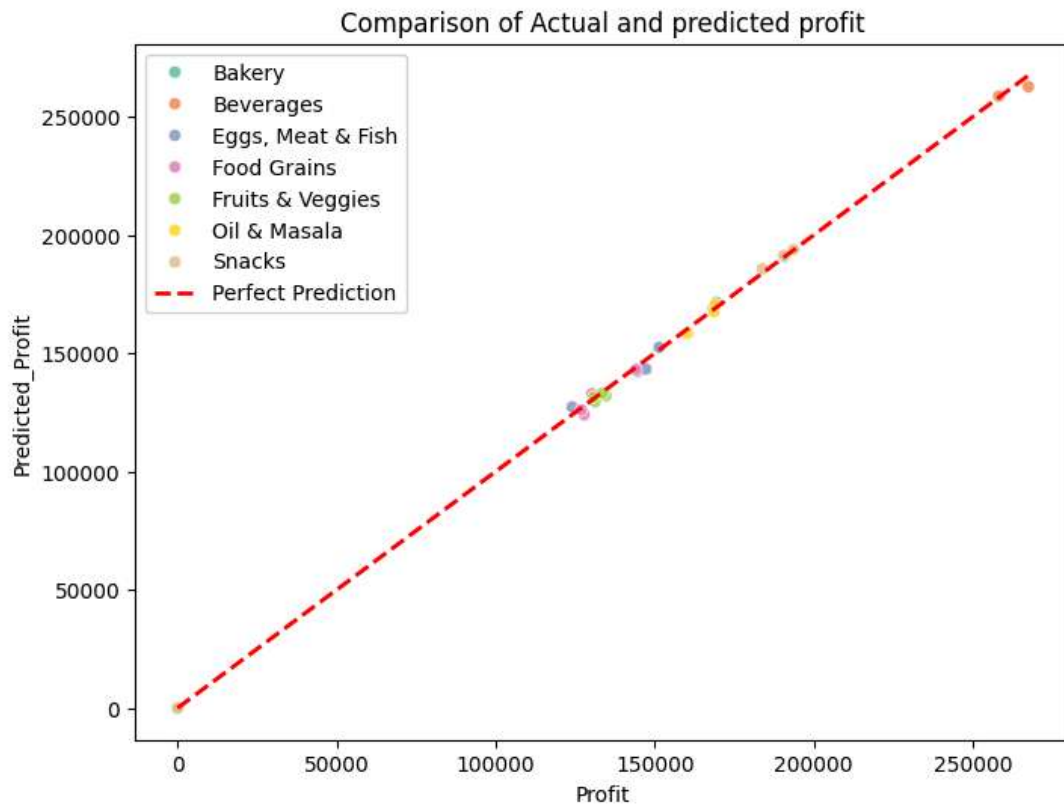


Fig 7.6: Comparison of Actual vs Predicted Profit

Forecasting 2019 Sales and Profit:

The trained model was used to forecast 2019 sales and profit based on data from 2015–2018.

The predicted results for selected categories and subcategories are shown below.

-----Next Year Predicted Sales & Profit-----				
	Category	Sub Category	Predicted_Sales	Predicted_Profit
1	Bakery	Biscuits	6.567697e+05	165393.431419
2	Bakery	Breads & Buns	7.608101e+05	199351.076978
3	Bakery	Cakes	6.844984e+05	167939.335876
36	Beverages	Health Drinks	1.059610e+06	257865.598646
44	Beverages	Soft Drinks	1.090035e+06	266574.594622
50	Eggs, Meat & Fish	Chicken	4.811335e+05	115165.403430
55	Eggs, Meat & Fish	Eggs	5.850361e+05	136843.589573
56	Eggs, Meat & Fish	Fish	5.759961e+05	159522.690680
61	Eggs, Meat & Fish	Mutton	6.148049e+05	152734.352029
69	Food Grains	Atta & Flour	5.125244e+05	126331.259724
76	Food Grains	Dals & Pulses	5.149107e+05	121351.787666
87	Food Grains	Organic Staples	5.596745e+05	139426.895877
89	Food Grains	Rice	5.151879e+05	127873.604434
103	Fruits & Veggies	Fresh Fruits	5.689499e+05	149163.043093
104	Fruits & Veggies	Fresh Vegetables	5.446437e+05	131683.397608
109	Fruits & Veggies	Organic Fruits	4.980892e+05	135202.138140
111	Fruits & Veggies	Organic Vegetables	4.986166e+05	124312.071973
123	Oil & Masala	Edible Oil & Ghee	6.249880e+05	154046.798742

Fig 7.7: Prediction Result of Year 2019

Highlights from Forecast Output:

- Bakery Items such as Biscuits and Breads & Buns are expected to reach around ₹5.6–₹7.8 lakh in sales with a profit between ₹1.6–₹1.9 lakh.
- Beverage Subcategories like Health Drinks and Soft Drinks are projected to generate over ₹9–₹10 lakh in sales, maintaining strong profitability.
- Eggs, Meat & Fish continue to show stable sales with good profit margins.
- Oil & Masala items have slightly lower projected sales but steady profit, suggesting consistent demand.

7.4 Dashboard Output and Prediction Interface

1. Analysis Tab

The Analysis Tab provides an overview of DMart's grocery data, allowing users to explore different analysis types such as Sales by City, Sales by Category, and Sales by Year.

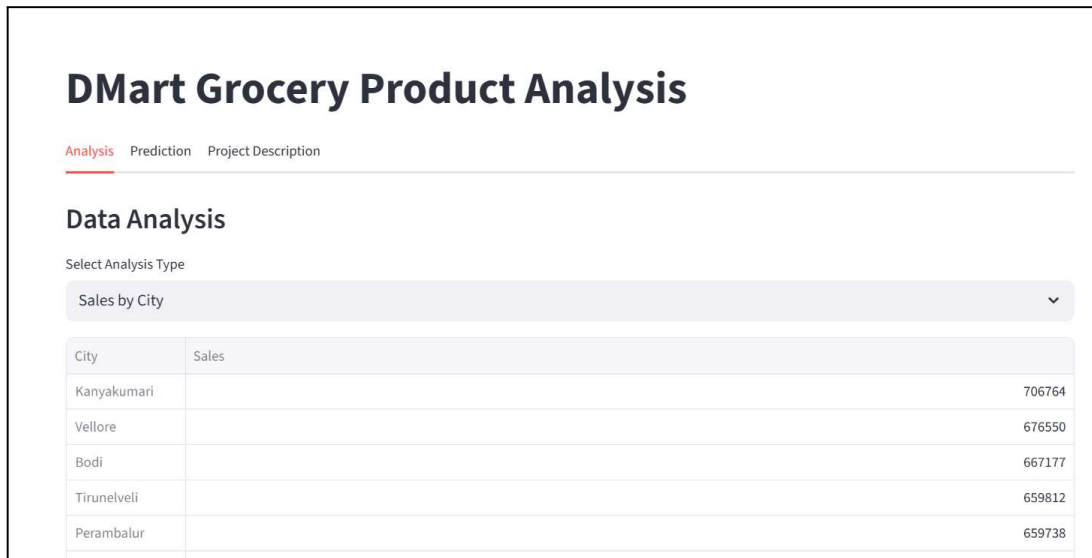


Fig 7.8: Analysis Tab Output

2. Prediction Tab

The Prediction Tab enables users to select a Category and Year to generate future predictions of sales and profit using the trained Random Forest model.

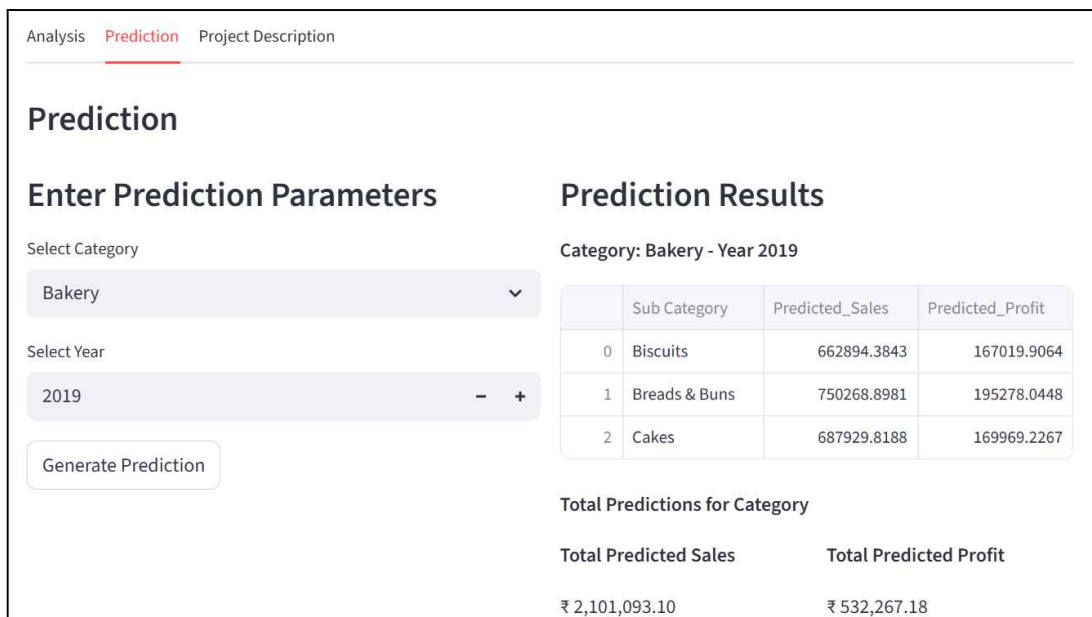


Fig 7.9: Prediction Tab Output

8 CONCLUSION AND DISCUSSION

8.1 Conclusion

The DMart Grocery Sales and Profit Prediction project successfully demonstrated the practical use of Machine Learning in retail business analytics. By using the Random Forest Regression model, the project accurately predicted future sales and profit trends based on past data (2015–2018).

Through exploratory data analysis (EDA), several useful insights were discovered — such as the top-performing product categories (Snacks, Beverages, Bakery), the highest revenue-generating regions (Urban Tamil Nadu cities), and the correlation between sales and profit. The predictive model provided strong performance, as seen in the close alignment between actual and predicted results.

Additionally, the development of an interactive Flask-based dashboard made the project more practical and user-friendly. The interface allowed users to:

- Analyze city-wise and category-wise sales data.
- Generate real-time sales and profit predictions for future years.
- Visualize performance trends using graphs and charts.

Overall, the project achieved its primary goal to design a reliable and interpretable system that supports data-driven decision-making for retail business operations. The integration of analytics and machine learning can help DMart and similar retail companies optimize inventory, pricing, and marketing strategies effectively.

8.2 Key Achievements

- Successfully implemented a Random Forest Regression model with high prediction accuracy.
- Discovered meaningful trends and relationships using EDA and correlation analysis.
- Built a user-interactive Flask web dashboard for visualizing and predicting grocery sales.

- Generated actionable business insights for profit improvement and operational planning.

8.3 Limitations

While the model and dashboard worked effectively, a few limitations were identified:

- The dataset did not include external factors like seasonal demand, competitor data, or marketing impact.
- Predictions were based only on past sales patterns and did not account for unforeseen market changes.
- The current model is limited to offline data and does not support real-time data updates.

9 REFERENCES

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